

COMPUTER SYSTEM ORGANIZATION

Time Allowed: 3 Hours

Full

Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten):
 $2 \times 10 = 20$

- i. The instruction, MOV AX, [2500H] is an example of

a) immediate addressing mode.

b) direct addressing mode.

c) indirect addressing mode.

d) register addressing mode.

00H] is an example of Explanation
The instruction indicates 'Moving' the contents of memory location at address '2500H' into the register AX, directly.

- ii. In a virtual memory system, which of the following can improve the efficiency of address translation?

a) Larger page size

b) Smaller page size

c) ~~Use of Translation Lookaside Buffer (TLB)~~

d) Increased RAM size

TLB is a hardware cache that stores recent transitions of virtual to physical memory addresses, significantly speeding up the address translation process.

- iii. Develop a simple assembly language program to add two numbers stored in registers AX and BX, and store the result in CX. Which of the following best represents the steps?

a) `MOV AX, BX; ADD CX, AX`

~~BY MOV CX, AX: ADD CX, BX~~

c) MOV AX, CX; ADD AX, BX

d) MOV BX, CX: ADD AX, CX

First, 'Moving' the content of 'AX' register to 'CX' register.

Second, 'Add' the content of 'BX' register with 'CX' register and finally store the

- iv. Which of the following is a challenge that arises in a RISC pipeline architecture?

a) Hazard detection and resolution

b) Fetching multiple instructions per cycle

c) Reducing the instruction set

- d) Supporting a large instruction set

What arises in a RISC pipeline architecture? *addition result at 'CX'.*

Q. 1. Different types of hazards like data hazard, control hazard, structural hazards are the main challenges of RISC pipeline.

- v. If a floating-point number uses 8 bits for the exponent and 23 bits for the mantissa, what is the precision limit for numbers represented by this format?

a) Approximately 16 decimal digits

b) Approximately 6-7 decimal digits

c) Approximately 12 decimal digits

d) Approximately 3-4 decimal digits

$$P_d = N_m \times \log_{10}(2)$$

↓ ↘
Precision digit No. of bits in Mantissa

$$P_d = 23 \times \log_{10}(2) \\ = 23 \times 0.30103 \approx 6.92 \approx 7$$

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$$\text{Hit ratio} = \frac{\text{No. of cache hits}}{(\text{Total memory references})}$$

↓
no. of hits + no. of misses

vi. What is the hit ratio in cache memory?

- a) Ratio of cache size to main memory size
- ☒ b) Number of cache hits divided by total memory accesses
- c) Ratio of cache misses to total memory accesses
- d) Ratio of cache speed to main memory speed

vii. In Intel 8086, which addressing mode uses the combination of a register and a displacement value?

- a) Immediate Addressing
- b) Direct Addressing
- c) Register Indirect Addressing
- ☒ d) Base-Indexed Addressing

In Base-Indexed Addressing mode the content of (Base Register + Index Register) and a displacement value is added.

viii. What is the main characteristic of the Von-Neumann architecture?

- a) Separate memory for data and instructions
- ☒ b) Shared memory for data and instructions
- c) No memory hierarchy
- d) Direct connection between CPU and memory

According to Von-Neumann architecture the memory is shared for both instruction and data.

ix. Which of the following is responsible for address sequencing in a microprogrammed control unit?

- a) Instruction Register
- b) Control Memory
- c) Arithmetic Logic Unit
- ☒ d) Program Counter

~~PC (Program Counter) is a register~~
Microprogram Sequencer (MS) is the register to store the current microinstruction address.

x. What is the word size of the Intel 8086 microprocessor?

- a) 8-bit
- ☒ b) 16-bit
- c) 32-bit
- d) 64-bit

The word size of Intel 8086 microprocessor is 16-bit.

xi. In a pipeline architecture, which stage comes immediately after Instruction Fetch?

- ☒ a) Decode
- b) Execute
- c) Memory Access
- d) Write Back

In pipeline architecture, the next stage after Instruction Fetch (IF) is Decode Instruction (DI).

xii. What is the function of cache memory in a computer system?

- a) To store the operating system
- ☒ b) To store frequently accessed data
- c) To increase the size of RAM
- d) To serve as backup storage

The function of cache memory is to store the frequently accessed data. The copy of RAM is stored in cache, which are accessed recently or frequently.



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- xiii. Which of the following micro-operations can combine two binary values?
a) Arithmetic micro-operation
b) Shift micro-operation
c) Logic micro-operation
d) Memory transfer
- In Arithmetic & Logic Microoperations we can combine two binary values, because for addition, subtraction, AND, OR, etc. arithmetical or logical operations a minimum two values are needed.*
- xiv. Which of the following is NOT a valid step in Booth's multiplication algorithm?
a) Add multiplicand
b) Shift right
c) Invert the result
d) Subtract multiplicand
- In Booth's Algorithm all the steps i.e. Add multiplicand, Subtract multiplicand, Shift right are there except Invert the result.*

GROUP-B

Answer any Five (05) questions.

2. Explain Von Neumann architecture. Discuss the various types of Addressing Modes of Intel 8086. 4+4
3. Explain Restoring Division Algorithm for Unsigned Integer. Multiply the (-9) with (-13) using booth's algorithm. 4+4
4. Explain 8086 internal architecture. Explain different types of interrupts. 5+3
5. Calculate the number of page faults for the following reference string with frame size as 3 by using a) LRU and FIFO page replacement algorithm. 5,0,2,1,0,3,0,2,4,3,0,3,2,1,3,0,1,5. 4+4
6. Compare between direct mapped, set associative and associative cache. Differentiate between write through and write back cache. 5+3
- *****
7. Differentiate between fixed point and floating-point representation. Represent -75 as 8-bit 2's complement number. What is Direct Memory Access(DMA)? 3+2+3
8. Explain Pipelining. Discuss different kinds of hazard occurs in pipelining. 3+5
9. Short Note (any two): 4+4
- a) Vector Processor
- b) RISC Pipeline
- c) Cache hit and cache miss

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COMPUTER SYSTEM ORGANISATION

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A is compulsory and to be answered first. This answer is to be made in separate loose script(s) provided for the purpose. Maximum time allowed is 30 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided. On early submission of answer scripts of Question No. 1, a student will get the remaining script earlier.

Answer any Five (05) Questions from Group B.

Group A

[1x20= 20]

1. Choose the correct alternative from the following (Any 20):

(i) Which of the following are the components of Von-Neumann architecture?

- (a) Busses, memory and input/output component,
- (b) Hard disks, floppy disks and CPU,
- (c) Memory, CPU and Printer,
- (d) Memory, input/output modules and CPU.

(ii) Which of the following computer memory is fastest?

- a) register
- b) Hard disk
- c) RAM
- d) None of the above.

(iii) In the parity generator which logic operator is used to generate the parity bit?

- (a) OR, (b) AND, (c) NOT, (d) XOR

(iv) Which of the following is not a characteristic of RISC?

- (a) One instruction per cycle
- (b) Large instruction set
- (c) Simple addressing modes
- (d) Register-to-register operations.

(v) What is computer organization?

- a) structure and behavior of a computer system as observed by the user
- b) structure of a computer system as observed by the developer
- c) structure and behavior of a computer system as observed by the developer
- d) All of the mentioned

(vi) The content of a 8 bit register is initially 10011100, what is the content of the register after an arithmetic shift-right operation?

- (a) 11001110, (b) 01001110, (c) 11001111, (d) 11001101.

(vii) The performance of cache memory is frequently measured in terms of

- a) main memory
- b) hit ratio
- c) CPU cycle
- d) Bus speed

(viii) Both the CISC and RISC architectures have been developed to reduce the _____

- a) Time delay
- b) Semantic gap
- c) Cost
- d) All of the mentioned

(ix) Which are the different type/s of generating control signals.

- a) Hardwired
- b) Micro-instruction
- c) Micro-programmed
- d) Both Micro-programmed and Hardwired

(x) The small extremely fast, RAM's are called as _____

- a) Heaps
- b) Accumulators
- c) Stacks
- d) Cache

(xi) MAR stands for _____

- (a) Memory Access Register,
- (b) Memory Address Register,
- (c) Memory Access Reference,
- (d) Memory Address Reference

(xii) In order to read multiple bytes of a row at the same time, we make use of _____

- a) Memory extension
- b) Cache
- c) Shift register
- d) Latch

(xiii) Cache memory uses _____

- (a) SRAM, (b) DRAM, (c) EEPROM, (d) EPROM.

(xiv) Computer address bus is _____

- a) Multidirectional
- b) Bidirectional
- c) Unidirectional
- d) None of the above

(xv) The address in the main memory is known as _____

- a) Logical address
- b) Physical address
- c) Virtual address
- d) None of the above

(xvi) Full form of RISC is

- a) Reduced instruction set computer
- b) Reduction instruction set computer
- c) Reduced instructor set computer
- d) none of these

(xvii) Which of the following statement(s) is(are) true?

- a) ROM is a read/write memory,
- b) PC points to the next instruction to be executed

- c) Stack works on the principle of FIFO,
- d) RAM is a Read/Write memory.

xviii) DMA stands for _____

- a) Direct memory adder
- b) Direct memory address
- c) Direct memory access
- d) none of these

xix) What is the full form of MBR

- a) memory buffer register
- b) memory boot register
- c) memory buffer reduction
- d) none of these

xx) IEEE double precision format is _____

- a) 30 bits
- b) 32 bits
- c) 60 bits
- d) 64 bits

xxi) Full form of PC is

- a) progress counter
- b) program count
- c) program counter
- d) none of these

xxii) Theoretically maximum speedup that a pipeline can provide depends on the _____

- a) number of segments on the pipeline
- b) number of Instructions
- c) number of address bus
- d) none of these

xxiii) Which memory is difficult to interface with processor

- a) Dynamic memory
- b) Static memory
- c) both a & b
- d) none of these

xxiv) An example of direct addressing mode is

- a) MOV C, B
- b) LDA 2021H
- c) both a & b
- d) none of these

xxv) Which mode of DMA is fastest?

- a) burst mode
- b) cycle stealing
- c) both a & b
- d) None of these

Group B

- 2.a. Draw the Von Neumann basic structure and mark all its components. 4
b. Write down the IEEE format for single and double precision numbers? 4
- 3.a. Write the restoring division process with flowchart and example. 5
b. Explain the function of BIU & EU. 3
4. a. What do you mean by hit and miss ratio? 4
b. A typical computer system cache memory access time is 8 ns and its main memory access time is 65 ns. If hit ratio is 90%, what is the average memory access time? 4
5. a. What do you mean by cachememory mapping? 2
b. Write down the direct cache memory mapping technique and what is its main disadvantage. 6
6. a. Write the differences between RISC and CISC architecture. 2
b. Discuss different types of addressing modes with examples. 6
7. a. Write the different pipeline hazards. 3
b. Explain the different groups of computers according to Flynn's classification. 5
- 8.a. Draw the pin diagram of 8086 microprocessor. 5
b. Explain maximum and minimum mode of 8086 microprocessor. 3
9. Write short notes on any two (a) TLB (b) DMA (c) Virtual Memory [4 + 4] 2
10. a. What do you mean by micro operation? 6
b. Explain the functioning procedure of micro-programmed control unit. 6

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Answer the following questions from Group-A, B & C as directed:

GROUP -A

1. Choose the correct alternative (Any ten)
 - i) The number successful accesses to memory stated as a fraction is called as
 - a) Access rate b) Success rate c) Hit rate d) Miss rate
 - ii) The final addition sum of the numbers, 0110 & 0110 is
 - a) 1101 b) 1111 c) 1001 d) 1010
 - iii) What does CSA stands for?
 - a) Computer Service Architecture b) Computer Speed Addition c) Carry Save Addition d) None of these
 - iv) Individual control word of the micro routine are called as-
 - a) Micro task b) Micro instruction c) Micro operation d) Micro Command
 - v) Which of the following circuit convert the binary data into a decimal?
 - a) Decoder b) Encoder c) Code-converter d) Multiplexer
 - vi) The situation wherein the data of operands are not available is called
 - a) Data hazard b) Stock c) Deadlock d) Structural hazard
 - vii) What is the full form of CISC?
 - a) Complex Instruction Sequential Compilation b) Complete Instruction Sequential Compilation c) Computer Integrated Sequential Compiler d) Complex Instruction Set Computer
 - viii) The alternate way of writing the instruction, ADD #5, R1 is-
 - a) ADD [5], [R1]; b) ADDI 5, R1; c) ADDIME 5, [R1]; d) There is no other way
 - ix) In order to read multiple bytes of a row at the same time, we make use of
 - a) Memory extension b) Cache c) Shift register d) Latch
 - x) In full adders the sum circuit is implemented using
 - a) And & or gates b) NAND gate c) XOR d) XNOR
 - xi) Computer address bus is
 - a) Unidirectional b) Bidirectional c) Multidirectional d) None of the above
 - xii) Which of the following computer bus connects the CPU to a memory on the system board?
 - a) Expansion bus b) Width bus c) System bus d) None of the above
 - xiii) The instructions that are used for reading an input port and writing an output port respectively are
 - a) MOV, XCHG, b) MOV, IN, c) IN, MOV, d) IN, OUT
 - xiv) Micro operation is shown as:
 - a) $R1 \leftarrow R2$ b) $R1 + R2$ c) Both d) None

xv) An interrupt that can be temporarily ignored is-
a) Vectored interrupt b) Non-maskable interrupt c) Maskable interrupt d) High priority interrupt

2. Fill in the blanks (Any ten):

- i) _____ is used to store data, instructions and results permanently for future use.
- ii) _____ is generally used to increase the apparent size of physical memory.
- iii) Gray Code is also called as _____.
- iv) Instruction register stores _____.
- v) A high speed memory is placed between the CPU and the primary memory is known as _____.
- vi) I/O address in 8086 is _____ bit.
- vii) Techniques that automatically move programs and data blocks into the physical memory when they are required for execution are called _____.
- viii) Hit ratio is maximum in _____ mapping.
- ix) The bias value for single-precision floating point numbers is _____.
- x) MOV AX, [2A50] is an example of _____ addressing mode.
- xi) Loop unrolling is a technique to improve _____.
- xii) Page table resides in _____.
- xiii) Microinstruction consists of _____.
- xiv) The smallest entity of memory is called _____.
- xv) A source program is usually in _____ language.

3. Answer the following question (any ten).

- i) How control unit controls other units?
- ii) Give an example of a 4 bit, 8bit, 16-bit, and 32 bit microprocessor.
- iii) What is Bus?
- iv) What is MAR and MDR?
- v) What is register?
- vi) What is interrupt?
- vii) What is non-volatile memory?
- viii) What is logical address?
- ix) Which is an error-detecting code?
- x) What is the logic shift?
- xi) What type of device converts digital signal into a form that is intelligible to the user?
- xii) Which memory stores instruction which is required to start a computer?
- xiii) Define clock rate.
- xiv) What is the RAID system?

GROUP -B

4. Answer the questions (Any six)

- i) What are the three main elements of the control unit?
- ii) What is Cache memory?
- iii) What is control memory address?
- iv) What is the 2's complement representation of -6?
- v) What is clock signal in COA?
- vi) Is USB is a bus?
- vii) Draw the block diagram of the half adder.
- viii) Draw a multiplication circuit diagram.
- ix) What's the difference between interrupt service routine and subroutine?
- x) What do you mean by the write-back policy?
- xi) What is RISC Pipeline?
- xii) What size of MUXs are needed?

GROUP -C

5. Answer the question (any one):

6x1

- a) Explain the components of the Computer system and what is micro operation?
- b) Represent $(12.625)_{10}$ in 32 bit floating point representation and what is odd parity checker?
- c) Describe the Von-Neumann Architecture with diagram? Explain the Bus Structure with examples

6. Answer the question (any one):

6x1

- a) Describe the Flag Register of 8086 microprocessor.
- b) Perform multiplication between 23 and 17 using fixed point multiplication algorithm.
- c) What are the key characteristics of micro-programmed control? Explain different types of micro operation.

7. Answer the question (any one):

6x1

- a) Discuss the various mapping techniques used in cache memory.
- b) What is virtual memory? How does it work?
- c) How can you interface RAM and the ROM EPROM to microprocessor 8086? What is the use of EPROM?

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In Question No. 1, out of 2 marks for each MCQ, 1 marks is allotted for right answer and 1 marks is allotted for correct explanation of the answer.
Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): $2 \times 10 = 20$

- i) Von Neumann architecture is based on _____
a) SISD b) SIMD c) MISD d) MIMD
- ii) Example for zero address instructions is _____
a) push b) load A c) move R1, A d) store x.
- iii) What is the full form of TLB ?
a) Translation Loop Buffer b) Translation Look-aside Buffer c) Time Loop Block d) None
- iv) What is Page Map Table ?
a) It maps the virtual addresses to physical addresses
b) It maps physical addresses to virtual addresses
c) It maps the virtual addresses to Logical addresses
d) Support all the above
- v) The result of MOV AL, 65 is to store _____
a) 0100 0010 in AL b) 42H in AL c) 40H in AL d) 0100 0001 in AL
- vi) Which of the following is page fault?
a. Page fault occurs when a program accesses a page of another program
b. Page fault occurs when a program accesses a page in main memory
c. Page fault occurs when there is an error in particular page
d. Page fault occurs when a program accesses a page which is not present in main memory
- vii) Which of the following are the two main components of the CPU?
a) CU and registers b) Registers and main memory c) CU and ALU d) Registers and ALU.
- viii) The term that provides simultaneous data processing tasks are _____
a) parallel processing b) array processing c) vector processing d) distributed processing.
- ix) A 16×8 Organisation of memory cells, can store upto _____
a) 256 bits b) 1024 bits c) 512 bits d) 128 bits
- x) A processor can access a memory location by 32 bits. Then find the total memory size if all memory locations are available to the processor,
a. 4 GB
b. 4 Gb
c. 2 GB
d. 4 MB
- xi) Data transfer from Cache Memory to Processor is _____
a) Word by Word b) Block by Block c) Block by Word d) Word by Block

xii) In which of the following term the performance of cache memory is measured?

- a. Chart ratio
- b. Hit ratio
- c. Cache ratio
- d. Data ratio

xiii) Which of these is NOT involved in the case of a memory write operation?

- a) Data bus b) MDR c) MAR d) PC

xiv) Which of the following memory unit communicates directly with the CPU?

- a. Auxiliary memory
- b. Main memory
- c. Secondary memory
- d. None of the above

xv) A computer system supports 2^{46} logical addresses and 2^{11} addresses per page. How many pages can be represented in secondary memory (virtual memory address space)?

- a) 35 b) 2^{35} c) 2^{25} d) 2^{57}

GROUP-B

Answer any Five (05) questions.

2. a) Draw the Von Neumann basic structure and mark all its components.

b) Draw and explain the BUS architecture of a digital computer?

3. a) Explain the basic instructions cycle with appropriate diagram.

b) Write the Zero address instructions to evaluate the arithmetic statement $X = (A+B) * (C+D)$

4. a) Write the difference between Minimum Mode and Maximum Mode in 8086 Microprocessor.

b) Explain with example any four addressing modes available in 8086 microprocessors.

5. a) Write the different cache mapping techniques and explain it.

b) A typical computer system cache memory access time is 8 ns and its main memory access time is 80 ns. If hit ratio is 90%, what is the average memory access time?

6. a) Write the difference between CISC & RISC Architecture.

b) Write short notes on any one: i) DMA ii) Virtual Memory

7. a) What is biased exponent of floating-point number?

b) Discuss in detail the architecture of 8086 microprocessor along with pin configuration diagram.

8. a) Explain the different groups of computers according to Flynn's classification.

b) Write down the IEEE-754 format for single and double precision numbers?

9. a) Describe the concept of Pipeline and its types?

b) Write the different types of pipeline hazards.